

FoodPro Detailed Database Structure

This document summarizes the current FoodPro database design, including relational tables from server/sql/init.sql and the JSONB-backed operational schema from server/entities.js. It highlights primary keys, important foreign keys, entity uniqueness rules, and project/location scoping behavior.

1. Hybrid Database Architecture

- FoodPro uses a hybrid PostgreSQL model.
- Relational tables store authentication, POS integration, procurement, goods receipt, and invoice flows.
- entity_records stores many operational modules as JSONB documents with server-side validation, uniqueness rules, and permission enforcement.
- The primary relational tables are ideal for multi-table transactional workflows, while entity_records provides flexibility for fast-changing business modules.

2. Relational Table Structure

users

Purpose: Application user master for authentication, role assignment, and project/location scoping.

Primary key: id

Important columns

- id TEXT PK
- email TEXT UNIQUE
- full_name TEXT
- role TEXT
- status TEXT
- site_id TEXT
- site_name TEXT
- password_hash TEXT
- temporary_password TEXT
- profile JSONB
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Relationships

- Referenced by auth_tokens.user_id
- Referenced by app_logs.user_id

auth_tokens

Purpose: Session token store for authenticated user access.

Primary key: token

Important columns

- token TEXT PK
- user_id TEXT FK

- created_at TIMESTAMPTZ
- expires_at TIMESTAMPTZ

Foreign keys

- user_id -> users.id ON DELETE CASCADE

Relationships

- Many auth tokens can belong to one user

entity_records

Purpose: Hybrid JSONB entity store for flexible business modules such as Inventory, Production, FoodWaste, Site, Recipe, and MaterialRequest.

Primary key: id

Important columns

- id TEXT PK
- entity_name TEXT
- data JSONB
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Relationships

- Acts as the main backing store for operational entities defined in server/entities.js
- Indexed by entity_name and selected JSONB paths for Inventory and InventoryLot

app_logs

Purpose: Application activity and page visit log.

Primary key: id

Important columns

- id TEXT PK
- user_id TEXT FK
- user_email TEXT
- page_name TEXT
- payload JSONB
- visited_at TIMESTAMPTZ

Foreign keys

- user_id -> users.id ON DELETE SET NULL

Relationships

- Many log rows can belong to one user

email_logs

Purpose: Outbound email dispatch tracking.

Primary key: id

Important columns

- id TEXT PK

- recipient TEXT
- subject TEXT
- payload JSONB
- status TEXT
- created_at TIMESTAMPTZ

pos_sources

Purpose: POS provider configuration including API credentials, sync frequency, and default project/site context.

Primary key: id

Important columns

- id TEXT PK
- name TEXT
- source_type TEXT
- api_url TEXT
- api_key TEXT
- api_secret TEXT
- sync_frequency TEXT
- is_active BOOLEAN
- default_site_id TEXT
- default_site_name TEXT
- settings JSONB
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Relationships

- Parent of pos_sales_orders
- Parent of pos_recipe_mapping
- Parent of pos_sync_logs

pos_sales_orders

Purpose: Imported POS order headers.

Primary key: id

Important columns

- id TEXT PK
- source_id TEXT FK
- external_order_id TEXT
- order_number TEXT
- site_id TEXT
- site_name TEXT
- location_name TEXT
- business_date DATE

- sold_at TIMESTAMPTZ
- currency TEXT
- total_amount NUMERIC(14,2)
- sync_method TEXT
- inventory_applied BOOLEAN
- raw_payload JSONB
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Foreign keys

- source_id -> pos_sources.id ON DELETE SET NULL

Relationships

- Parent of pos_sales_items
- Unique external order key on (source_id, external_order_id)

pos_sales_items

Purpose: Imported POS order line items that can map to FoodPro recipes for deduction and variance analysis.

Primary key: id

Important columns

- id TEXT PK
- order_id TEXT FK
- external_item_id TEXT
- pos_item_code TEXT
- pos_item_name TEXT
- recipe_id TEXT
- recipe_name TEXT
- quantity NUMERIC(14,3)
- unit_price NUMERIC(14,2)
- total_price NUMERIC(14,2)
- site_id TEXT
- site_name TEXT
- deduction_status TEXT
- raw_payload JSONB
- created_at TIMESTAMPTZ

Foreign keys

- order_id -> pos_sales_orders.id ON DELETE CASCADE

Relationships

- Many sales items belong to one POS sales order

pos_recipe_mapping

Purpose: Mapping between POS menu items and FoodPro recipes.

Primary key: id

Important columns

- id TEXT PK
- source_id TEXT FK
- pos_item_code TEXT
- pos_item_name TEXT
- recipe_id TEXT
- recipe_name TEXT
- servings_per_sale NUMERIC(14,3)
- site_scope TEXT
- site_id TEXT
- auto_deduct BOOLEAN
- notes TEXT
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Foreign keys

- source_id -> pos_sources.id ON DELETE CASCADE

Relationships

- Lookup table used during POS import and inventory deduction

pos_sync_logs

Purpose: Execution log for POS syncs and uploads.

Primary key: id

Important columns

- id TEXT PK
- source_id TEXT FK
- sync_type TEXT
- status TEXT
- started_at TIMESTAMPTZ
- finished_at TIMESTAMPTZ
- records_received INTEGER
- records_imported INTEGER
- records_skipped INTEGER
- message TEXT
- request_payload JSONB
- response_payload JSONB
- created_at TIMESTAMPTZ

Foreign keys

- source_id -> pos_sources.id ON DELETE SET NULL

Relationships

- Many sync logs can belong to one POS source

suppliers

Purpose: Supplier master used in procurement and invoice tracking.

Primary key: id

Important columns

- id TEXT PK
- name TEXT
- contact_person TEXT
- email TEXT
- phone TEXT
- address TEXT
- city TEXT
- country TEXT
- payment_terms TEXT
- lead_time_days INTEGER
- status TEXT
- rating NUMERIC(6,2)
- categories JSONB
- notes TEXT
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Relationships

- Parent of purchase_orders
- Parent of goods_receipts
- Parent of supplier_invoices
- Parent of supplier_price_history

purchase_requests

Purpose: Procurement request header from manual requests or low-stock automation.

Primary key: id

Important columns

- id TEXT PK
- request_number TEXT UNIQUE
- site_id TEXT
- site_name TEXT
- request_date DATE

- needed_by DATE
- requested_by TEXT
- requested_by_name TEXT
- priority TEXT
- status TEXT
- approval_role TEXT
- approved_by TEXT
- approved_by_name TEXT
- approved_at TIMESTAMPTZ
- auto_generated BOOLEAN
- source_type TEXT
- notes TEXT
- total_estimated_cost NUMERIC(14,2)
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Relationships

- Parent of purchase_request_items
- Can be referenced by purchase_orders

purchase_request_items

Purpose: Line items for purchase requests.

Primary key: id

Important columns

- id TEXT PK
- request_id TEXT FK
- ingredient_id TEXT
- ingredient_name TEXT
- description TEXT
- requested_quantity NUMERIC(14,3)
- approved_quantity NUMERIC(14,3)
- ordered_quantity NUMERIC(14,3)
- unit TEXT
- estimated_unit_price NUMERIC(14,2)
- line_total NUMERIC(14,2)
- status TEXT
- preferred_supplier_id TEXT
- preferred_supplier_name TEXT
- created_at TIMESTAMPTZ

Foreign keys

- request_id -> purchase_requests.id ON DELETE CASCADE

Relationships

- Many request items belong to one purchase request
- Can be referenced by purchase_order_items

purchase_orders

Purpose: Purchase order header issued against a request and supplier.

Primary key: id

Important columns

- id TEXT PK
- po_number TEXT UNIQUE
- request_id TEXT FK
- supplier_id TEXT FK
- supplier_name TEXT
- site_id TEXT
- site_name TEXT
- order_date DATE
- expected_delivery_date DATE
- currency TEXT
- subtotal NUMERIC(14,2)
- tax_amount NUMERIC(14,2)
- total_amount NUMERIC(14,2)
- status TEXT
- approved_by TEXT
- approved_by_name TEXT
- approved_at TIMESTAMPTZ
- created_by TEXT
- created_by_name TEXT
- received_percentage NUMERIC(8,2)
- notes TEXT
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Foreign keys

- request_id -> purchase_requests.id ON DELETE SET NULL
- supplier_id -> suppliers.id ON DELETE SET NULL

Relationships

- Parent of purchase_order_items
- Parent of goods_receipts
- Referenced by supplier_invoices

purchase_order_items

Purpose: Line items on purchase orders.

Primary key: id

Important columns

- id TEXT PK
- order_id TEXT FK
- request_item_id TEXT FK
- ingredient_id TEXT
- ingredient_name TEXT
- ordered_quantity NUMERIC(14,3)
- received_quantity NUMERIC(14,3)
- unit TEXT
- unit_price NUMERIC(14,2)
- line_total NUMERIC(14,2)
- status TEXT
- created_at TIMESTAMPTZ

Foreign keys

- order_id -> purchase_orders.id ON DELETE CASCADE
- request_item_id -> purchase_request_items.id ON DELETE SET NULL

Relationships

- Many items belong to one purchase order
- Can be referenced by goods_receipt_items
- Can be referenced by supplier_price_history

goods_receipts

Purpose: Goods receipt note header for received purchase orders.

Primary key: id

Important columns

- id TEXT PK
- grn_number TEXT UNIQUE
- purchase_order_id TEXT FK
- supplier_id TEXT FK
- supplier_name TEXT
- site_id TEXT
- site_name TEXT
- receipt_date DATE
- received_by TEXT
- received_by_name TEXT
- status TEXT

- notes TEXT
- created_at TIMESTAMPTZ

Foreign keys

- purchase_order_id -> purchase_orders.id ON DELETE CASCADE
- supplier_id -> suppliers.id ON DELETE SET NULL

Relationships

- Parent of goods_receipt_items
- Can be referenced by supplier_invoices

goods_receipt_items

Purpose: Receipt lines with accepted/rejected quantities, batch numbers, and expiry dates.

Primary key: id

Important columns

- id TEXT PK
- receipt_id TEXT FK
- order_item_id TEXT FK
- ingredient_id TEXT
- ingredient_name TEXT
- received_quantity NUMERIC(14,3)
- accepted_quantity NUMERIC(14,3)
- rejected_quantity NUMERIC(14,3)
- unit TEXT
- batch_number TEXT
- expiry_date DATE
- status TEXT
- created_at TIMESTAMPTZ

Foreign keys

- receipt_id -> goods_receipts.id ON DELETE CASCADE
- order_item_id -> purchase_order_items.id ON DELETE SET NULL

Relationships

- Many receipt items belong to one goods receipt
- Used to populate InventoryLot and InventoryTransaction in app logic

supplier_invoices

Purpose: Supplier invoice header tied to procurement activity.

Primary key: id

Important columns

- id TEXT PK
- invoice_number TEXT UNIQUE
- supplier_id TEXT FK

- supplier_name TEXT
- purchase_order_id TEXT FK
- goods_receipt_id TEXT FK
- invoice_date DATE
- due_date DATE
- subtotal NUMERIC(14,2)
- tax_amount NUMERIC(14,2)
- total_amount NUMERIC(14,2)
- status TEXT
- entered_by TEXT
- entered_by_name TEXT
- notes TEXT
- created_at TIMESTAMPTZ
- updated_at TIMESTAMPTZ

Foreign keys

- supplier_id -> suppliers.id ON DELETE SET NULL
- purchase_order_id -> purchase_orders.id ON DELETE SET NULL
- goods_receipt_id -> goods_receipts.id ON DELETE SET NULL

Relationships

- Many invoices can be associated with procurement flows

supplier_price_history

Purpose: Historical ingredient price by supplier and purchase order line.

Primary key: id

Important columns

- id TEXT PK
- supplier_id TEXT FK
- supplier_name TEXT
- ingredient_id TEXT
- ingredient_name TEXT
- purchase_order_item_id TEXT FK
- unit_price NUMERIC(14,2)
- currency TEXT
- effective_date DATE
- lead_time_days INTEGER
- site_id TEXT
- site_name TEXT
- created_at TIMESTAMPTZ

Foreign keys

- supplier_id -> suppliers.id ON DELETE CASCADE
- purchase_order_item_id -> purchase_order_items.id ON DELETE SET NULL

Relationships

- Used by supplier variance and procurement comparison reports

3. Relationship Summary

- users (1) -> auth_tokens (many) by auth_tokens.user_id
- users (1) -> app_logs (many) by app_logs.user_id
- pos_sources (1) -> pos_sales_orders (many) by pos_sales_orders.source_id
- pos_sales_orders (1) -> pos_sales_items (many) by pos_sales_items.order_id
- pos_sources (1) -> pos_recipe_mapping (many) by pos_recipe_mapping.source_id
- pos_sources (1) -> pos_sync_logs (many) by pos_sync_logs.source_id
- purchase_requests (1) -> purchase_request_items (many) by purchase_request_items.request_id
- purchase_requests (1) -> purchase_orders (many/optional) by purchase_orders.request_id
- suppliers (1) -> purchase_orders (many) by purchase_orders.supplier_id
- purchase_orders (1) -> purchase_order_items (many) by purchase_order_items.order_id
- purchase_orders (1) -> goods_receipts (many) by goods_receipts.purchase_order_id
- goods_receipts (1) -> goods_receipt_items (many) by goods_receipt_items.receipt_id
- purchase_order_items (1) -> goods_receipt_items (many/optional) by goods_receipt_items.order_item_id
- suppliers (1) -> supplier_invoices (many) by supplier_invoices.supplier_id
- purchase_orders (1) -> supplier_invoices (many/optional) by supplier_invoices.purchase_order_id
- goods_receipts (1) -> supplier_invoices (many/optional) by supplier_invoices.goods_receipt_id
- suppliers (1) -> supplier_price_history (many) by supplier_price_history.supplier_id
- purchase_order_items (1) -> supplier_price_history (many/optional) by supplier_price_history.purchase_order_item_id
- entity_records is the parent store for JSONB-backed business entities such as Site, User, Ingredient, Recipe, Inventory, Production, MaterialRequest, and FoodWaste

4. entity_records Schema Layer

- Site: Project hierarchy master. Unique by name and project_code. Drives Company -> Region -> Location -> Kitchen -> Store/Warehouse.
- User: Operational user profile. Unique by email. Stores role, site_id, allowed_site_ids, visibility_scope, and role_permissions.
- RoleProfile: Custom role definition. Unique by role_key and name. Stores access_level and permissions.
- Ingredient: Ingredient master. Unique by name, ingredient_code, and sku.
- Recipe: Recipe master. Unique by name and recipe_code.
- Inventory: Current stock by site and ingredient. Unique by site_id + ingredient_id.
- InventoryLot: Batch/lot balance by ingredient, site, batch_number, expiry_date.
- InventoryTransaction: Movement ledger for receipt, issue, production, transfer, and adjustment.
- MenuPlan: Planned recipe/meal schedule by project/site/date.

- Production: Production request / batch planning and execution record.
- MaterialRequest: Project/site material request linked to approved production and procurement flow.
- FoodWaste: Waste posting with ingredient/recipe/batch linkage, cost, and approval status.
- WasteTarget: Waste reduction targets by project/site and month.
- ProductionTransfer: Inter-location finished goods movement.
- AdvancedReportSchedule: Saved report schedule definitions.
- ERPIntegrationConfig: ERP provider connection and field mapping settings.
- ERPIntegrationLog: ERP export history and retry tracking.
- ForecastScenario: Demand forecasting scenario definition.
- ForecastSnapshot: Calculated forecast output snapshot.
- AttendanceSession / AttendanceRecord / StaffShift: Labor scheduling and attendance entities.
- CustomerMealPlan / BranchOrder / QualityControl / QRCode / QRDelivery / RFQ: Supporting operational modules built on the hybrid entity model.

5. Key Uniqueness and Integrity Rules

- Ingredient: unique by name; ingredient_code and sku are also unique when provided.
- Recipe: unique by name; recipe_code is unique when provided.
- Site: unique by project name; project_code is unique when provided.
- User: unique by email.
- RoleProfile: unique by role_key and role name.
- UserGroup: unique by group name.
- Inventory: unique by site_id + ingredient_id, which allows only one current stock record per item per location.

6. Indexes and Performance Notes

- idx_entity_records_entity_name on entity_records(entity_name)
- idx_entity_records_entity_updated_at on entity_records(entity_name, updated_at DESC)
- idx_pos_sales_orders_source_external UNIQUE on pos_sales_orders(source_id, external_order_id) when external_order_id is not null
- idx_pos_recipe_mapping_lookup on pos_recipe_mapping(source_id, pos_item_code, pos_item_name)
- idx_purchase_requests_status on purchase_requests(status, request_date DESC)
- idx_purchase_orders_status on purchase_orders(status, order_date DESC)
- idx_goods_receipts_order on goods_receipts(purchase_order_id, receipt_date DESC)
- idx_supplier_invoices_supplier on supplier_invoices(supplier_id, invoice_date DESC)
- idx_supplier_price_history_lookup on supplier_price_history(ingredient_id, supplier_id, effective_date DESC)
- idx_entity_records_inventory_lookup on entity_records((data->>'site_id'), (data->>'ingredient_id')) for Inventory
- idx_entity_records_inventory_lot_lookup on entity_records((data->>'site_id'), (data->>'ingredient_id'), (data->>'batch_number')) for InventoryLot

7. Project / Location Scoping

- Server-side scoping is applied for Site, User, Inventory, InventoryLot, InventoryTransaction, Production, ProductionBatch, ProductionTransfer, MenuPlan, FoodWaste, MaterialRequest, Recipe, Attendance, Forecast, ERP logs, WasteTarget, and related operational entities.
- User assignment uses `site_id`, `allowed_site_ids`, and `visibility_scope`.
- For non-admin users, only assigned projects/locations or their permitted subtree are visible.
- This means data relationships exist globally in the database, but access to rows is filtered by project/site scope in application logic.

8. Practical Reading Guide

- If you need raw auth/session structure, start with `users` and `auth_tokens`.
- If you need procurement chain tracing, follow `purchase_requests` -> `purchase_request_items` -> `purchase_orders` -> `purchase_order_items` -> `goods_receipts` -> `goods_receipt_items` -> `supplier_invoices`.
- If you need POS chain tracing, follow `pos_sources` -> `pos_sales_orders` -> `pos_sales_items`, with `pos_recipe_mapping` used during recipe linkage.
- If you need operational module data such as Production, MaterialRequest, FoodWaste, Inventory, or Site, read `entity_records` together with `entityRegistry` in `server/entities.js`.
- If you need to understand why a user sees or does not see project data, inspect `locationScope.js` and the fields `site_id`, `allowed_site_ids`, and `visibility_scope` on the User entity.